



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 3

1. Title: Implement control structures and form validation. Create a grading system based on user-entered marks.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

A. Form Validation

Ensures user input is accurate and valid before processing.

- **Empty Check:** Ensures the input field is not blank.
- **Type Check:** Ensures input is a number using `isNaN()`.
- **Range Check:** Restricts marks within valid bounds.

B. Control Structures (**if...else if...else**)

Executes specific blocks of code depending on whether conditions evaluate to **true** or **false**. Used to map mark ranges to grades.

C. DOM Manipulation

- **Input:** `document.getElementById('marks').value` (Reads input).
- **Output:** `document.getElementById('result').innerText` (Displays output).



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4. Output:

Sonal's Student Grading System (24070521199):

Enter Marks:

Marks = 100
Grade = A

5. Result/Conclusion: By using control structures (`if...else if`) alongside form validation techniques, web applications can accurately process user inputs, prevent faulty data processing, and produce dynamic, real-time results such as student grading reports.



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Case Study No.: 3

1. Case Study Title: Write a JavaScript program that accepts student marks and displays the scholarship category.

2. Case Study Program Code:

```
<> StudentGrade.html • <> Scholarship.html X
<> Scholarship.html > html > body > script > checkEligibility
1  <!DOCTYPE html>           //Sonal Nakhate 24070521199
2  <html>
3      <head>
4          <title> Scholarship Eligibility </title>
5      </head>
6      <body>
7          <h1> Student Scholarship Eligibility Check:</h1>
8          Enter marks:
9          <input type = "number" id = "marks">
10         <button onclick = "checkEligibility()">Submit</button>
11         <p id = "result"></p>
12         <script>
13             function checkEligibility()
14             {
15                 var marks = document.getElementById("marks").value;
16                 if (marks == "")
17                 {
18                     alert("Please Enter Marks: ");
19                     return;
20                 }
21                 if (marks< 0 || marks> 100)
22                 {
23                     alert("Enter marks between 0 and 100");
24                     return;
25                 }
26                 if (marks>=95)
27                     sch = "Platinum Scholarship!";
28                 else if (marks >= 85)
29                     sch = "Gold Scholarship!";
30                 else if (marks >= 70)
31                     sch = "Silver Scholarship!";
32                 else if (marks>=50)
```



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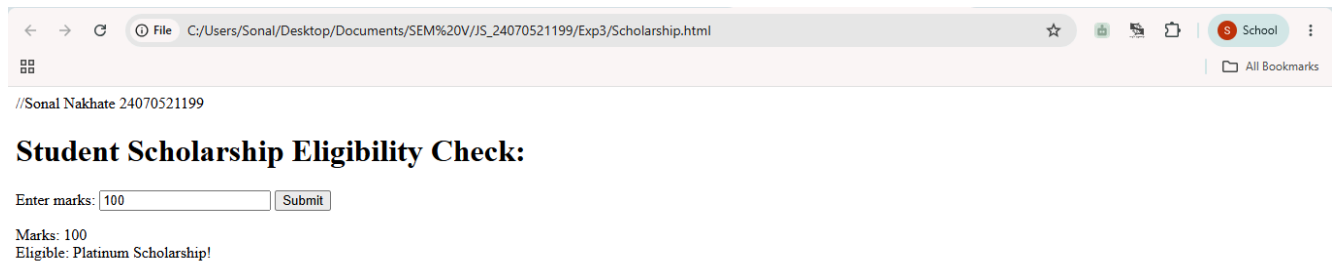
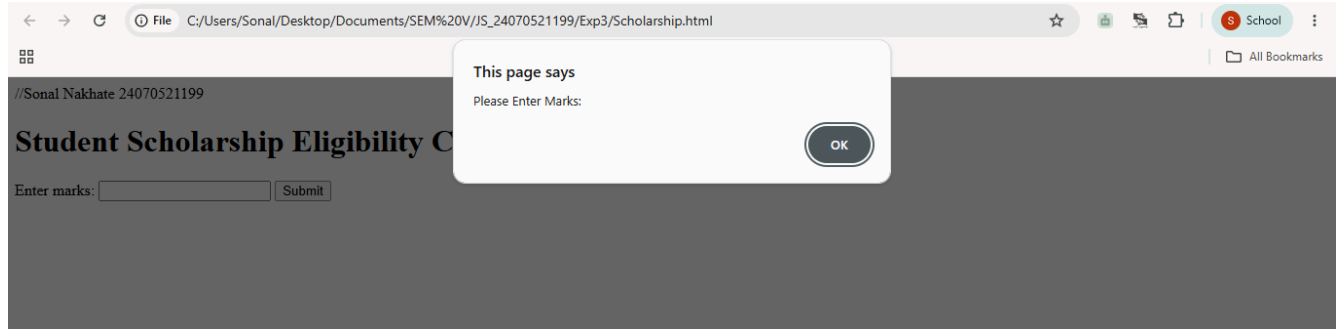
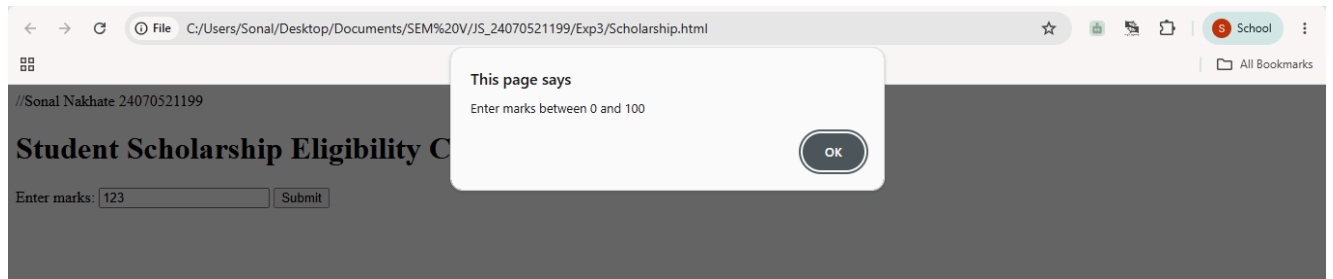
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```
30     else if (marks >= 70)
31     sch = "Silver Scholarship!";
32     else if (marks>=50)
33     sch = "Bronze Scholarship!";
34     else
35     sch = "Not Eligible";
36     document.getElementById("result").innerHTML = "Marks: " + marks + "<br>
37     }
38     </script>
39     </body>
40 </html>
```

3. Output:





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4. Result/Conclusion: We have successfully implemented a simple JavaScript program that checks scholarship eligibility based on marks scored by the student.